Decoherence and Open Quantum Systems

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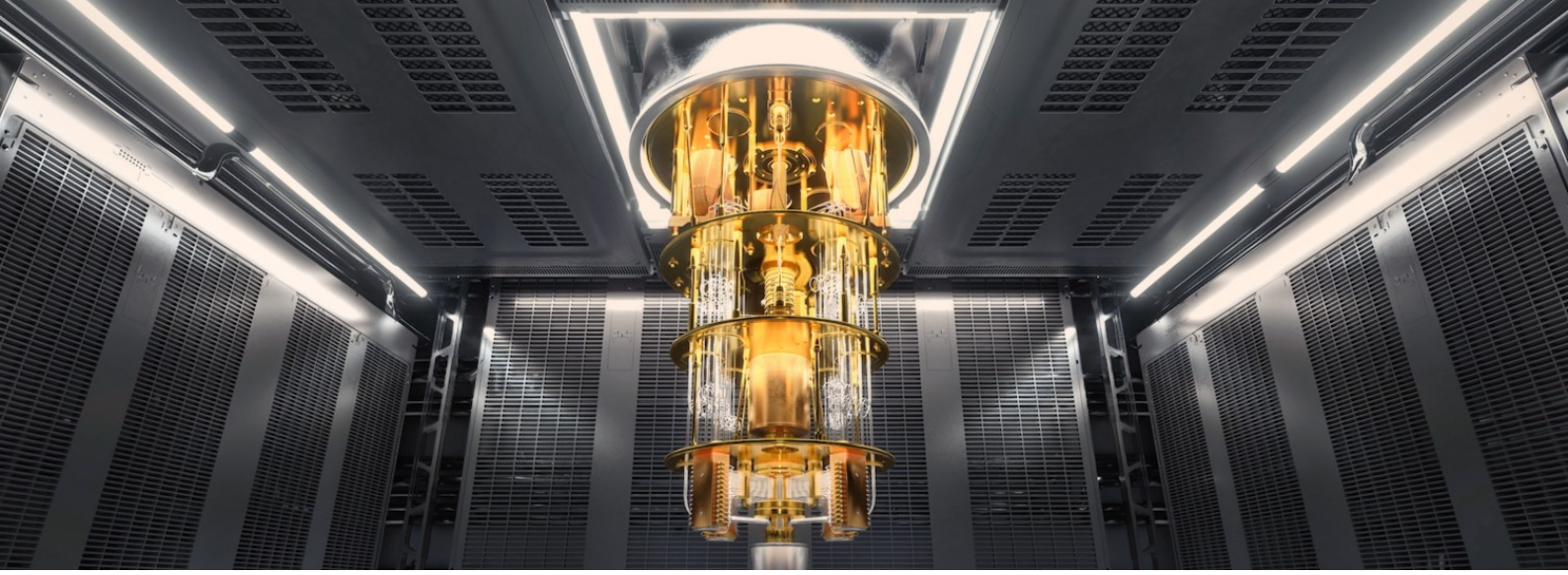

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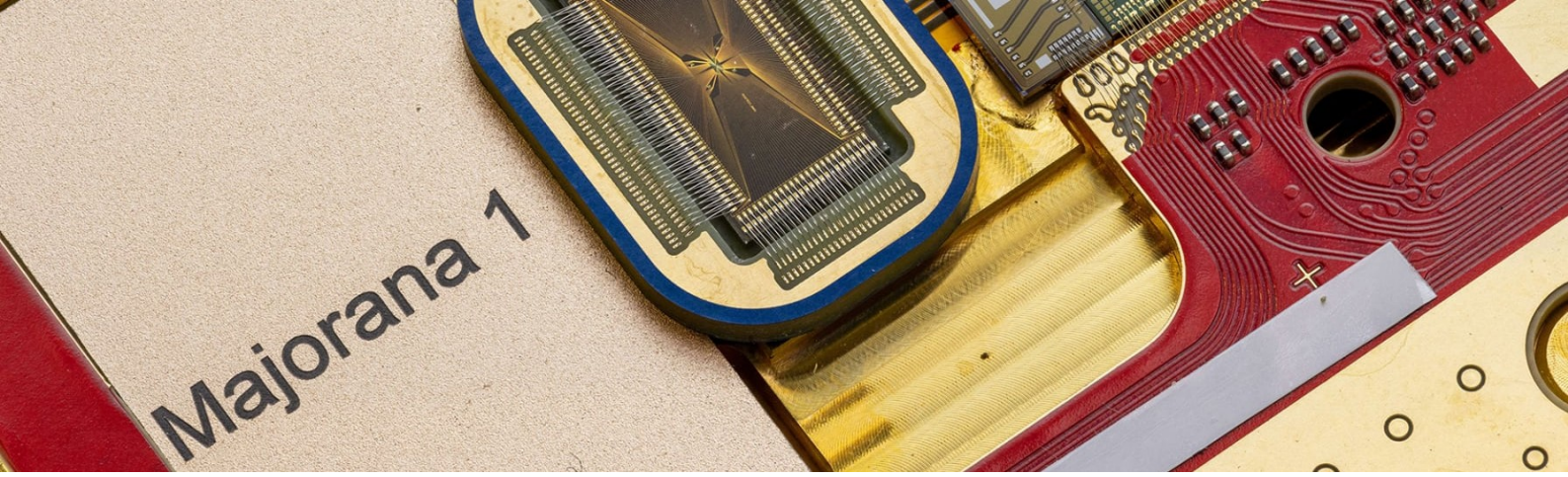


1. Mathematical Preliminaries

The theory of decoherence and Open Quantum Systems is mathematically rich, requiring a solid foundation in functional analysis, probability theory, stochastic calculus, linear algebra, operator theory, and Gaussian processes. Unlike closed quantum systems, where evolution is unitary and governed by the Schrödinger equation, an open system interacts with an environment typically infinite dimensional, noisy, stochastic, and uncontrollable. To treat such systems rigorously, we begin by developing mathematical machinery that underpins density matrices, correlation functions, noise processes, master equations, and quantum dynamical maps.

In this chapter we review these tools in a detailed, rigorous and physically motivated way, preparing the ground for master equations, Lindblad form, Kraus operators, quantum channels and decoherence mechanisms in later chapters. The treatment here is designed to be slow, explanatory and self-contained, relying mainly on classical QM, Probability, Statistical Mechanics and Gaussian Stochastic Process formulations from *Breuer & Petruccione* and our course notes.

1.1 Linear Algebra Foundations

We begin with the algebraic structure of quantum states. At the heart of quantum theory lies the concept of a complex vector space with an inner product a Hilbert space.

Definition 1.1.1 — Vector Space. A set V is called a vector space over a field $\mathbb{C}$ if for any $u, v \in V$ and $\alpha, \beta \in \mathbb{C}$,

$$\alpha u + \beta v \in V,$$

and the usual associative, distributive properties hold.

Quantum states are elements of such vector spaces. But more structure is needed.

Definition 1.1.2 — Inner Product. An inner product on V is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{C}$ satisfying:

$$\langle u, v \rangle = \langle v, u \rangle^*, \tag{1.1}$$

$$\langle u, u \rangle \geq 0, \qquad \langle u, u \rangle = 0 \iff u = 0, \tag{1.2}$$

$$\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle. \tag{1.3}$$

Definition 1.1.3 — Hilbert Space. A Hilbert space $\mathscr{H}$ is a complete inner product space over $\mathbb{C}$. Completeness means all Cauchy sequences converge inside $\mathscr{H}$.

R Completeness is not a technical formality: it ensures that limits of physically meaningful sequences (wavepacket spreads, operator approximations) remain within the space of valid states.

1.1.1 Cayley–Hamilton Theorem

The Cayley–Hamilton theorem is a basic result in linear algebra that will be useful when we deal with matrix functions (such as e^{At}), spectral decompositions, and evolution operators for finite-dimensional systems.

Definition 1.1.4 — Characteristic Polynomial. Let A be an $n \times n$ matrix. Its characteristic polynomial is

$$p_A(\lambda) = \det(\lambda I - A) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0,$$

where the c_k are scalars (depending on A).

Theorem 1.1.1 — Cayley–Hamilton. Every square matrix satisfies its own characteristic polynomial:

$$p_A(A) \equiv A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I = 0.$$

In other words, the matrix A is a root of its characteristic polynomial when we substitute $\lambda \mapsto A$.

R [Use in Matrix Functions] For any polynomial or analytic function $f(A)$ (such as e^{At}), the Cayley–Hamilton theorem allows us to reduce $f(A)$ to a linear combination of $\{I, A, \ldots, A^{n-1}\}$. This is particularly useful for 2×2 and 3×3 matrices (Bloch-equation type generators, finite-level Hamiltonians, etc.) where the dynamics can be expressed in closed form.

Special cases and simple examples.

■ **Example 1.1 — Diagonal Matrices.** If A is diagonal,

$$A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n),$$

then

$$p_A(\lambda) = \prod_{k=1}^{n} (\lambda - \lambda_k).$$

Substituting A gives

$$p_A(A) = \prod_{k=1}^{n} (A - \lambda_k I) = \operatorname{diag}\big(p_A(\lambda_1), \ldots, p_A(\lambda_n)\big) = 0,$$

because each diagonal entry $p_A(\lambda_k) = 0$. This shows the theorem immediately in the diagonal case and also gives intuition for the general case. ■

■ **Example 1.2 — 2×2 Case.** For a 2×2 matrix,

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the characteristic polynomial is

$$p_A(\lambda) = \lambda^2 - (\mathrm{tr}A)\lambda + \det A.$$

Cayley–Hamilton says

$$A^2 - (\mathrm{tr}A)A + (\det A)I = 0.$$

Thus any power A^n with $n \geq 2$ can be expressed as a linear combination of I and A. In particular, one can write

$$e^{At} = \alpha(t)I + \beta(t)A$$

for suitable scalar functions $\alpha(t)$ and $\beta(t)$, which is often used in solving two-level system dynamics and Bloch-equationtype problems. ■

R In later chapters, when we exponentiate finite-dimensional Hamiltonians or Lindblad generators, the Cayley–Hamilton theorem will allow us to reduce matrix exponentials and other functions of A to a finite basis of powers of A. This is especially convenient for two-level systems and 2×2 Bloch-type evolution matrices.

1.1.2 Bras, kets and operators

In Dirac notation, vectors are kets $|\psi\rangle$ and linear functionals are bras $\langle\phi|$. The inner product is $\langle\phi|\psi\rangle$. Operators $\hat{A} : \mathscr{H} \to \mathscr{H}$ act linearly.

Definition 1.1.5 — Adjoint. For every operator $\hat{A}$, its adjoint $\hat{A}^\dagger$ satisfies

$$\langle\phi|\hat{A}|\psi\rangle = \langle\psi|\hat{A}^\dagger|\phi\rangle^*.$$

Definition 1.1.6 — Hermitian Operator. $\hat{A} = \hat{A}^\dagger$. These represent observables.

Definition 1.1.7 — Unitary Operator. $\hat{U}^\dagger\hat{U} = I$. These represent reversible evolutions.

■ **Example 1.3** Time evolution for a closed quantum system is unitary:

$$|\psi(t)\rangle = e^{-iHt}|\psi(0)\rangle.$$

■

R In open system dynamics, time evolution is non-unitary. This will lead us to completely positive trace-preserving maps later.

1.2 Density Matrices and Mixed States

Pure states $|\psi\rangle$ describe maximal information, but open systems naturally evolve into mixtures. The density operator plays a central role.

Definition 1.2.1 — Density Matrix. A density operator ρ is a positive semidefinite operator with unit trace:

$$\rho \geq 0, \qquad \mathrm{Tr}(\rho) = 1.$$

Theorem 1.2.1 — Spectral Decomposition of Density Matrices. Every density matrix can be written as

$$\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i|, \qquad p_i \geq 0, \quad \sum_i p_i = 1.$$

R A pure state has $\rho^2 = \rho$, entropy zero. Mixed states satisfy $\rho^2 \neq \rho$, entropy > 0.

Expectation of observable $\hat{A}$:

$$\langle A \rangle = \mathrm{Tr}(\rho A).$$

This trace rule generalizes $\langle\psi|A|\psi\rangle$ and naturally extends to ensembles and decoherence situations.

Summary and relation to state vectors.

For a system prepared in a definite pure state $|\psi\rangle$, the density operator is simply $\rho = |\psi\rangle\langle\psi|$, and expectation values reduce to the familiar formula $\langle A \rangle = \langle\psi|A|\psi\rangle$. The usefulness of ρ becomes clear when the preparation is not perfectly known: if the system is prepared as $|\psi_i\rangle$ with probability p_i, then the correct object that reproduces all measurement statistics is the average operator $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$. In this sense the density matrix is the quantum analogue of a classical probability distribution: instead of assigning probabilities to points in phase space, it assigns weights to projectors onto quantum states. All observable predictions are obtained from ρ alone via $\langle A \rangle = \mathrm{Tr}(\rho A)$, so the density matrix may be regarded as the "state" of the system in the operational sense. Pure states correspond to projectors with $\rho^2 = \rho$, while mixed states represent statistical uncertainty about which $|\psi_i\rangle$ was prepared. This unified description allows both pure and probabilistic preparations to be handled within the same framework and explains why the density operator is the natural generalization of the state vector in quantum mechanics.

1.3 Operators, Spectrum, Trace and Positivity

Definition 1.3.1 — Spectrum. The spectrum $\sigma(A)$ of operator A is the set $\{\lambda \in \mathbb{C} \mid A - \lambda I \text{ non-invertible}\}$.

Theorem 1.3.1 — Spectral Theorem. Every Hermitian operator has a real spectrum and orthonormal eigenbasis.

$$A = \sum_i \lambda_i |i\rangle \langle i|.$$

This is crucial for quantum systems because observables return real measurement outcomes.

Definition 1.3.2 — Trace.

$$\mathrm{Tr}(A) = \sum_i \langle i|A|i\rangle,$$

basis-independent.

R Trace positivity ensures probabilities are non-negative.

Examples will appear later in stochastic master equations.

1.4 Probability Theory Foundations

Open systems interact with environments modelled by randomness.

Definition 1.4.1 — Probability Space. A probability space is $(\Omega, \mathscr{F}, P)$, where Ω = sample space, $\mathscr{F}$ = σ-algebra, P = probability measure.

Definition 1.4.2 — Random Variable. A function $X : \Omega \to \mathbb{R}$ measurable w.r.t. $\mathscr{F}$.

$$\mathbb{E}[X] = \int_\Omega X \, dP$$

Expectation generalizes to operators later via trace.

Definition 1.4.3 — Gaussian Random Variable. X is Gaussian if its distribution is

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}.$$

R Gaussianity is central to decoherence models such as CaldeiraLeggett bath and quantum Brownian motion.

1.5 Stochastic Processes and Gaussian Processes

Definition 1.5.1 — Stochastic Process. A family $\{X(t)\}_{t\in\mathbb{R}}$ of random variables on probability space $(\Omega, \mathscr{F}, P)$.

Definition 1.5.2 — Gaussian Process. A process is Gaussian if every finite set $(X(t_1), \ldots, X(t_n))$ has joint Gaussian distribution.

Correlation function:

$$C(t,s) = \mathbb{E}[X(t)X(s)].$$

Breuer & Petruccione emphasize Gaussian processes as models for thermal environments, correlation noise and master equations.

■ **Example 1.4 — OrnsteinUhlenbeck Noise.**

$$\dot{X}(t) = -\gamma X(t) + \eta(t),$$

where η is white noise. Leads to exponential correlation decay. ■

R Quantum bath correlation functions determine decoherence rate via master equation kernels.

1.6 Why All This Matters for Decoherence

Open systems require:

- Hilbert space $\otimes$ environment space structure
- density matrices for mixed states
- stochastic processes for noise environments
- trace over bath degrees of freedom
- operator positivity and completely positive maps

Everything above is needed to understand the master equation derivation in forthcoming chapters.

1.7 Tensor Product Structure of Composite Systems

Interaction with an environment means the total Hilbert space is

$$\mathscr{H}_{\text{tot}} = \mathscr{H}_S \otimes \mathscr{H}_E,$$

where S is the system and E the environment (bath, reservoir, field modes, phonons, etc.).

Definition 1.7.1 — Tensor Product of Hilbert Spaces. For $|\psi\rangle \in \mathscr{H}_S$ and $|\phi\rangle \in \mathscr{H}_E$, their composite state is $|\psi\rangle \otimes |\phi\rangle$.

R Tensor product allows correlations and entanglement. Unlike classical probability, quantum subsystems cannot generally be described by separate states they may not factorize.

1.7.1 Reduced State and Partial Trace

The environment contains many degrees of freedom not directly observed; we trace them out.

Definition 1.7.2 — Partial Trace. For ρ_{SE} on $\mathscr{H}_S \otimes \mathscr{H}_E$,

$$\rho_S = \mathrm{Tr}_E(\rho_{SE}), \qquad \rho_E = \mathrm{Tr}_S(\rho_{SE}).$$

■ **Example 1.5** If $\rho_{SE} = \rho_S \otimes \rho_E$, tracing over E gives ρ_S unchanged. But if there is entanglement, ρ_S becomes mixed. ■

R Tracing over the environment generates decoherence and dissipation because inaccessible degrees of freedom act as noise sources.

1.7.2 Schmidt Decomposition

Theorem 1.7.1 — Schmidt Decomposition. Any pure bipartite state $|\Psi\rangle \in \mathscr{H}_S \otimes \mathscr{H}_E$ can be written as

$$|\Psi\rangle = \sum_i \sqrt{\lambda_i}\,|s_i\rangle\,|e_i\rangle\,, \qquad \lambda_i \geq 0, \quad \sum_i \lambda_i = 1.$$

R Decoherence arises when system states $|s_i\rangle$ become entangled with distinguishable environment states $|e_i\rangle$. Off-diagonal terms vanish when $\langle e_i|e_j|e_i|e_j\rangle \approx \delta_{ij}$.

1.7.3 Von Neumann Entropy and Mixedness

Definition 1.7.3 — Von Neumann Entropy.

$$S(\rho) = -\mathrm{Tr}(\rho \log \rho).$$

- $S = 0$ for pure states
- $S > 0$ for mixed states
- Decoherence increases entropy of ρ_S

■ **Example 1.6** Maximally mixed state in dimension d:

$$\rho = \frac{I}{d}, \qquad S = \log d.$$

■

R Entropy growth is a signature of open system dynamics. In closed systems it remains constant under unitary evolution.

1.8 Probability Measures and Moments

We now go deeper into probability tools needed for stochastic baths.

Definition 1.8.1 — Characteristic Function.

$$\phi_X(k) = \mathbb{E}[e^{ikX}].$$

Expanding:

$$\phi_X(k) = 1 + ik\mathbb{E}[X] - \frac{k^2}{2}\mathbb{E}[X^2] + \cdots$$

Definition 1.8.2 — Moments and Cumulants. nth moment: $m_n = \mathbb{E}[X^n]$
nth cumulant: κ_n where

$$\log \phi_X(k) = \sum_{n=1}^{\infty} \frac{(ik)^n}{n!}\kappa_n.$$

R Gaussian variables have only first two cumulants nonzero. This makes Gaussian noise analytically tractable in quantum Brownian motion.

1.9 Correlation Functions and Noise Characterization

Definition 1.9.1 — Autocorrelation Function.

$$C(\tau) = \mathbb{E}[X(t)X(t+\tau)].$$

R Autocorrelation encodes memory. Markovian processes decay rapidly, non-Markovian retain long memory causing long-time coherence tails.

■ **Example 1.7 — Exponential Correlation.**

$$C(\tau) = \sigma^2 e^{-\gamma|\tau|}.$$

Short correlation = Markovian limit. ■

Noise spectral density is Fourier transform of correlation.

Definition 1.9.2 — Power Spectral Density (Wiener-Khinchin).

$$S(\omega) = \int_{-\infty}^{\infty} d\tau\, e^{-i\omega\tau} C(\tau).$$

Theorem 1.9.1 — Wiener-Khinchin Theorem. Autocorrelation and spectral density form a Fourier pair.

R Decoherence rate is determined by overlap between system frequency and environmental spectral density $S(\omega)$.

1.10 White Noise and Gaussian Processes

White noise $\eta(t)$:

$$\mathbb{E}[\eta(t)] = 0, \qquad \mathbb{E}[\eta(t)\eta(t')] = 2D\delta(t-t').$$

Definition 1.10.1 — Brownian Motion.

$$B(t) = \int_0^t \eta(s)\, ds.$$

R Quantum Brownian motion is the prototype for decoherence of position superpositions in harmonic oscillator baths.

Gaussian processes arise from central limit behavior of baths.

Theorem 1.10.1 — Gaussian Process Characterization. $X(t)$ Gaussian iff every linear combination $\sum_i \alpha_i X(t_i)$ is Gaussian.

■ **Example 1.8** Thermal bath operator $B(t)$ in CaldeiraLeggett model is Gaussian with correlation $\propto \coth(\beta\omega/2)$. ■

1.11 Linear Operators As Random Variables

Quantum measurements sample operator eigenvalues statistically.

Definition 1.11.1 — Quantum Expectation.

$$\langle A \rangle = \mathrm{Tr}(\rho A).$$

Definition 1.11.2 — Variance.

$$\mathrm{Var}(A) = \langle A^2 \rangle - \langle A \rangle^2.$$

R Random environments lead to operator-valued stochasticity and noise averaging often appears inside traces.

1.12 Quantum-Classical Correspondence in Noise

We distinguish:

1. **Classical noise:** random external field 2. **Quantum noise:**

Proposition 1.12.1 Quantum noise comes from bath operators

$$H_{SE} = \sum_{\alpha} A_\alpha \otimes B_\alpha.$$

R Tracing out bath yields stochastic influence functional, generating non-unitary system dynamics.

1.13 Master Equation Motivation Preview

For weak coupling:

$$\dot{\rho}_S = -i[H_S, \rho_S] + \mathscr{D}[\rho_S],$$

dissipator $\mathscr{D}$ depends on bath correlations.

- Markovian: exponential decay; Lindblad equation emerges
- Non-Markovian: memory kernels; integro-differential evolution

R All formal tools developed in this chapter will support rigorous derivation of such master equations later.

1.14 Dynamical Maps and Open System Evolution

For closed systems we have unitary evolution

$$\rho(t) = U(t)\rho(0)U^\dagger(t),$$

but for open systems interacting with a bath,

$$\rho_S(t) = \mathrm{Tr}_E \left(U_{SE}(t)\, \rho_{SE}(0)\, U_{SE}^\dagger(t) \right),$$

which is generally non-unitary and may increase entropy.

Definition 1.14.1 — Quantum Dynamical Map. A linear map Λ_t such that

$$\rho_S(t) = \Lambda_t[\rho_S(0)].$$

We require:

1. **Trace Preservation:** probabilities sum to one 2. **Positivity:** density matrices stay positive 3. **Complete Positivity:** positivity holds even with ancilla extensions

Definition 1.14.2 — Completely Positive Trace-Preserving (CPTP) Map. Λ is CPTP if

$$\Lambda \otimes I_n \geq 0 \quad \forall n,$$

and $\mathrm{Tr}(\Lambda[\rho]) = 1$ for all ρ.

R Complete positivity is stricter than positivity. It prevents unphysical negative probabilities when system is entangled with another.

1.15 Kraus Representation of Quantum Channels

Theorem 1.15.1 — Kraus Representation. Any CPTP map can be written as

$$\Lambda[\rho] = \sum_k K_k \rho K_k^\dagger, \qquad \sum_k K_k^\dagger K_k = I.$$

Sketch. Start with dilation theorem:

$$\Lambda[\rho] = \mathrm{Tr}_E \left(U(\rho \otimes |0\rangle\langle 0|) U^\dagger \right).$$

Expand in an environment basis $\{|e_k\rangle\}$ to obtain Kraus operators $K_k = \langle e_k | U | 0 \rangle$. ■

R Kraus operators represent measurement-like interactions of the system with its environment. They will later describe decoherence channels.

1.15.1 Examples of Quantum Channels (Preview)

■ **Example 1.9 — Phase Damping Channel.**

$$K_0 = \sqrt{1-p} I, \qquad K_1 = \sqrt{p} |0\rangle\langle 0|, \qquad K_2 = \sqrt{p} |1\rangle\langle 1|.$$

Destroys coherence without energy exchange. ■

■ **Example 1.10 — Amplitude Damping Channel.** Models spontaneous emission.

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}.$$

■

These will reappear in decoherence later.

1.16 Norms, Distances and Fidelity Between States

We need ways to quantify how much coherence is lost.

Definition 1.16.1 — Trace Distance.

$$D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1 = \frac{1}{2} \mathrm{Tr} \sqrt{(\rho - \sigma)^2}.$$

R Operational meaning: maximal probability difference in measurement outcomes.

Definition 1.16.2 — Fidelity.

$$F(\rho,\sigma) = \left(\mathrm{Tr}\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right)^2.$$

- $0 \le F \le 1$
- $F = 1$ iff $\rho = \sigma$
- $F < 1$ measures decoherence strength

1.17 Superoperators and Liouvillian Form

Definition 1.17.1 — Superoperator. A map acting on operators: $\mathscr{L} : \mathscr{B}(\mathscr{H}) \to \mathscr{B}(\mathscr{H})$.

In open dynamics we write

$$\dot{\rho}_S(t) = \mathscr{L}[\rho_S(t)].$$

Later, Lindblad form will be derived:

$$\mathscr{L}[\rho] = -i[H,\rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right).$$

R L_k Lindblad operators encode decoherence channels determined by environment correlation functions introduced earlier.

1.18 Measure Theory and Stochastic Integration (Overview)

To handle continuous noise:

Definition 1.18.1 — σ-Algebra. A set closed under countable unions and complements.

Definition 1.18.2 — Stochastic Integral.

$$\int_0^t X(s)\, dB(s)$$

defined via Ito calculus.

R Ito vs Stratonovich integration becomes relevant in quantum Langevin equations.

1.19 Putting It All Together Why This Chapter Matters

This chapter built the formal toolkit required for open quantum systems:

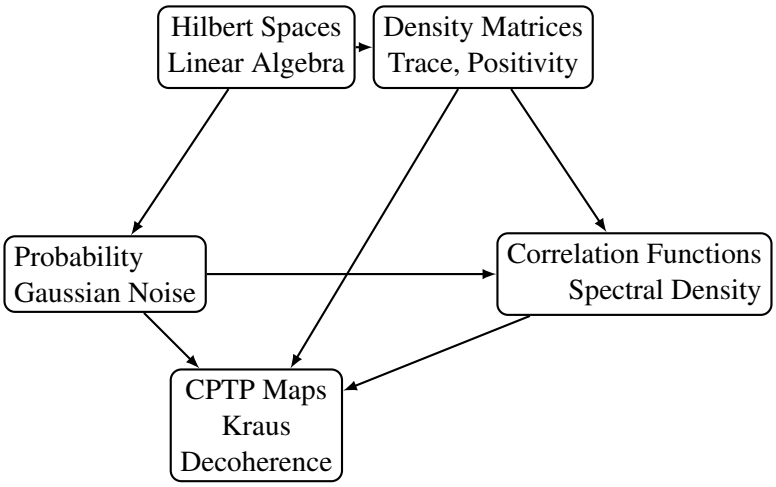
- Hilbert space, operators, density matrices
- Tensor products, partial trace, Schmidt decomposition
- Entropy, mixedness, decoherence interpretation
- Probability theory, Gaussian processes, stochastic noise
- Correlation functions, spectral densities (WienerKhinchin)
- Positive and completely positive maps
- CPTP channels and Kraus representation

All of these elements unify in master equations and decoherence theory. Without them, later results cannot be understood rigorously.

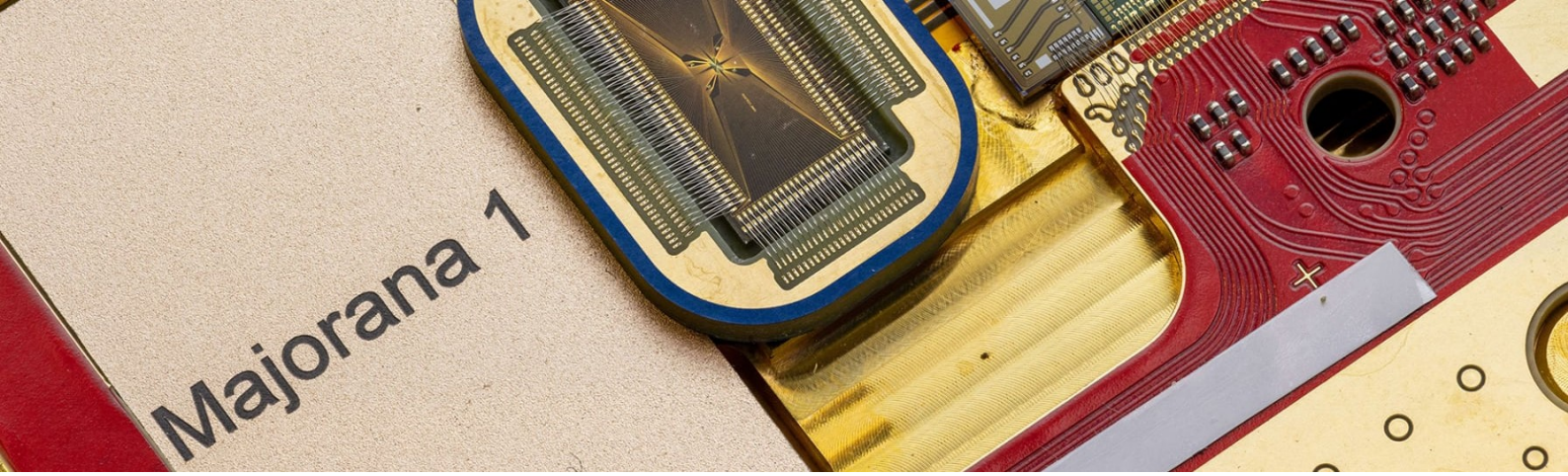
1.20 Exercises for the Reader

1. Prove that $\mathrm{Tr}_E(|\Psi\rangle\langle\Psi|)$ is mixed whenever Schmidt rank > 1.
2. For Gaussian noise with $C(\tau) = \sigma^2 e^{-\gamma|\tau|}$, compute $S(\omega)$ and discuss Markovian limit $\gamma \to \infty$.
3. Show that trace distance is contractive under CPTP maps.
4. Verify complete positivity of amplitude damping channel.
5. Compute entropy $S(\rho)$ for a qubit undergoing phase damping.
6. Show that fidelity increases under purification of states.

1.21 Visual Summary



All mathematical structures converge here leading directly to decoherence theory.



2. The Structure of Quantum States

In these notes on decoherence and open quantum systems, we will often move between several complementary ways of describing quantum states:

- **Hilbert space (ket / bra) picture:** states as vectors or density operators.
- **Phase-space picture:** states as quasi-probability distributions (Wigner, P, Q functions).
- **Operator-algebra picture:** states as positive trace-one operators, acted on by linear maps (quantum channels).

For open systems, these different viewpoints become essential: the environment is often modelled as a Gaussian bosonic bath in phase space, the reduced system state is a density matrix, and the evolution is captured by completely positive trace-preserving (CPTP) maps.

In this chapter we lay down the mathematical structure needed later:

1. Phase-space description and Wigner function.
2. Covariance matrix formalism and Gaussian states.
3. Positive maps, complete positivity and Kraus representation.
4. Schmidt decomposition and entanglement in bipartite systems.
5. Two-level systems and the Bloch sphere representation.

Standard references for much of this material are Breuer and Petruccione [1], Nielsen and Chuang [9], Gardiner and Zoller [5], Walls and Milburn [13], and lecture notes by Chaturvedi [3].

2.1 Phase-Space Description of Quantum States

Classical mechanics describes a particle by a point (q,p) in phase space. Probabilistic descriptions use a density $\rho(q,p) \geq 0$ with

$$\int dq\,dp\,\rho(q,p) = 1.$$

Observables are functions $f(q,p)$ and expectations are

$$\langle f \rangle_{\mathrm{cl}} = \int dq\,dp\,f(q,p)\rho(q,p).$$

Quantum mechanics cannot assign sharp values to q and p because they are non-commuting operators. Nevertheless, we can associate to every quantum state a function on phase space a *quasi*-probability distribution that reproduces classical-like expectation formulas.

2.1.1 Canonical Operators and Weyl Displacements

Definition 2.1.1 — Canonical Commutation Relations. The canonical position and momentum operators satisfy

$$[\hat{q}, \hat{p}] = i\hbar.$$

It is convenient to introduce the annihilation and creation operators

$$a = \sqrt{\frac{m\omega}{2\hbar}}\,\hat{q} + i\sqrt{\frac{1}{2m\omega\hbar}}\,\hat{p}, \qquad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\,\hat{q} - i\sqrt{\frac{1}{2m\omega\hbar}}\,\hat{p},$$

which satisfy $[a, a^\dagger] = 1$.

Definition 2.1.2 — Weyl Displacement Operator. For a complex number $\alpha \in \mathbb{C}$, the displacement operator is

$$D(\alpha) = \exp\left(\alpha a^\dagger - \alpha^* a\right).$$

These operators translate coherent states in phase space and form the basic building blocks of phase-space representations. They obey the Weyl commutation relation

$$D(\alpha)D(\beta) = e^{i\,\mathrm{Im}(\alpha\beta^*)}D(\alpha+\beta).$$

2.1.2 Wigner Function: Definition and Basic Properties

The Wigner function is the most widely used quasi-probability distribution for continuous-variable systems.

Definition 2.1.3 — Wigner Function. Let ρ be a density operator. The Wigner function $W(q,p)$ is defined by

$$W(q,p) = \frac{1}{2\pi\hbar}\int_{-\infty}^{\infty} dy\, e^{ipy/\hbar}\left\langle q - \frac{y}{2}\middle|\rho\middle|q + \frac{y}{2}\right\rangle.$$

Proposition 2.1.1 — Reality. The Wigner function $W(q,p)$ is real-valued.

Sketch. Taking complex conjugates and using the hermiticity of ρ, $\langle x|\rho|y\rangle^* = \langle y|\rho|x\rangle$, one finds $W(q,p)^* = W(q,p)$. ■

Proposition 2.1.2 — Normalization.

$$\int_{-\infty}^{\infty} dq \int_{-\infty}^{\infty} dp\, W(q,p) = 1.$$

Sketch. Insert the definition, integrate over p to get a delta function in y, then use $\mathrm{Tr}(\rho) = 1$. ■

Theorem 2.1.3 — Correct Marginals. Let $\rho = |\psi\rangle\langle\psi|$. Then

$$\int_{-\infty}^{\infty} dp\, W(q,p) = |\psi(q)|^2, \qquad \int_{-\infty}^{\infty} dq\, W(q,p) = |\phi(p)|^2,$$

where $\psi(q) = \langle q|\psi\rangle$ and $\phi(p) = \langle p|\psi\rangle$.

Sketch. For the first marginal, integrate over p,

$$\int dp\, e^{ipy/\hbar} = 2\pi\hbar\,\delta(y),$$

so only $y = 0$ contributes, giving $\langle q|\rho|q\rangle = |\psi(q)|^2$. The momentum marginal is similar in the momentum representation. ■

R Thus $W(q,p)$ is not a probability density (it can be negative), but it does reproduce the correct position and momentum distributions as marginals.

2.1.3 Coherent States and Their Wigner Function

Definition 2.1.4 — Coherent State. A coherent state $|\alpha\rangle$ is defined as the eigenstate of a:

$$a|\alpha\rangle = \alpha\,|\alpha\rangle, \qquad \alpha \in \mathbb{C}.$$

It can be generated by displacing the vacuum:

$$|\alpha\rangle = D(\alpha)|0\rangle.$$

Coherent states are minimum-uncertainty states with $\Delta q\,\Delta p = \hbar/2$, and they follow classical-like trajectories in phase space.

Theorem 2.1.4 — Wigner Function of a Coherent State. The Wigner function of a coherent state $|\alpha\rangle$ is a Gaussian in phase space centered at the classical point corresponding to α:

$$W_\alpha(q,p) = \frac{1}{\pi\hbar}\exp\left[-\frac{(q-q_0)^2}{\Delta q^2} - \frac{(p-p_0)^2}{\Delta p^2}\right],$$

where

$$q_0 = \sqrt{\frac{2\hbar}{2m\omega}}\,\mathrm{Re}(\alpha), \quad p_0 = \sqrt{2\hbar m\omega}\,\mathrm{Im}(\alpha),$$

and $\Delta q^2 = \frac{\hbar}{2m\omega}$, $\Delta p^2 = \frac{\hbar m\omega}{2}$.

Sketch. Compute the position-space wavefunction of a coherent state,

$$\psi_\alpha(q) = \langle q|\alpha\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\exp\left[-\frac{m\omega}{2\hbar}(q-q_0)^2 + \frac{i}{\hbar}p_0 q + \text{phase}\right],$$

insert into the Wigner definition and carry out the Gaussian integral over y or use the characteristic function approach (see, e.g., [13] or [5]). ■

R Coherent states behave most classically: their Wigner function is a positive Gaussian blob in phase space that moves along a classical trajectory for a harmonic oscillator. Decoherence of superpositions of coherent states (Schrödinger cat states) is often studied via the rapid suppression of interference fringes in their Wigner functions.

2.2 Covariance Matrices and Gaussian States

We now generalize the coherent state example to arbitrary Gaussian states of one or multiple modes, which are completely characterized by their first and second moments.

2.2.1 Quadratures and Phase-Space Vectors

For a single mode, define the quadrature operators

$$\hat{q} = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger), \qquad \hat{p} = i\sqrt{\frac{\hbar m\omega}{2}}(a^\dagger - a).$$

For n modes, assemble them into a $2n$-dimensional vector:

$$R = (\hat{q}_1, \hat{p}_1, \hat{q}_2, \hat{p}_2, \ldots, \hat{q}_n, \hat{p}_n)^T.$$

Definition 2.2.1 — Symplectic Form. The canonical commutation relations can be written compactly as

$$[R_j, R_k] = i\hbar\Omega_{jk},$$

where the $2n \times 2n$ symplectic matrix is

$$\Omega = \bigoplus_{k=1}^{n} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

2.2.2 Covariance Matrix and First Moments

Definition 2.2.2 — First Moments. The displacement vector d is

$$d_j = \langle R_j \rangle = \mathrm{Tr}(\rho R_j).$$

Definition 2.2.3 — Covariance Matrix. The covariance matrix σ has entries

$$\sigma_{jk} = \frac{1}{2}\langle R_j R_k + R_k R_j \rangle - \langle R_j \rangle \langle R_k \rangle.$$

Theorem 2.2.1 — RobertsonSchrödinger Uncertainty Principle. For any quantum state,

$$\sigma + i\frac{\hbar}{2}\Omega \geq 0,$$

as a matrix inequality.

Idea. The inequality follows by considering the expectation value of $(\xi \cdot R)^2$ for arbitrary complex vectors ξ and using the positivity of $\langle X^\dagger X \rangle$ for all operators X; see e.g. [14]. ■

R The matrix inequality encodes all uncertainty relations between all pairs of quadrature operators. For a single mode, it reduces to $(\Delta q)^2(\Delta p)^2 \geq \hbar^2/4$.

2.2.3 Gaussian States and Characteristic Functions

Definition 2.2.4 — Gaussian State. A state ρ is Gaussian if its characteristic function

$$\chi(\xi) = \mathrm{Tr}\big(\rho\, e^{i\xi^T \Omega R}\big)$$

is a Gaussian function of ξ, i.e. of the form

$$\chi(\xi) = \exp\Big(-\tfrac{1}{2}\xi^T \sigma \xi + i d^T \xi\Big),$$

for some covariance matrix σ and displacement vector d.

For Gaussian states, knowing σ and d is equivalent to knowing ρ, up to irrelevant global phases.

■ **Example 2.1 — Thermal State of a Harmonic Oscillator.** For $H = \hbar\omega(a^\dagger a + 1/2)$ and $\rho \propto e^{-\beta H}$, the state is Gaussian with zero mean $d = 0$. The covariance matrix is diagonal in (q, p) basis and encodes the BoseEinstein occupation number $\bar{n} = (e^{\beta\hbar\omega} - 1)^{-1}$. ■

2.2.4 Symplectic Transformations and Williamsons Theorem

Definition 2.2.5 — Symplectic Transformation. A real $2n \times 2n$ matrix S is symplectic if

$$S\Omega S^T = \Omega.$$

Such transformations correspond to linear canonical transformations in phase space.

Theorem 2.2.2 — Williamsons Theorem. For any covariance matrix σ, there exists a symplectic matrix S such that

$$\sigma = S^T \nu S,$$

where

$$\nu = \bigoplus_{k=1}^{n} \begin{pmatrix} \nu_k & 0 \\ 0 & \nu_k \end{pmatrix},$$

with $\nu_k \geq \hbar/2$. The ν_k are called symplectic eigenvalues of σ.

R Williamson decomposition is the Gaussian analogue of diagonalizing a covariance matrix. It reduces any Gaussian state to a product of thermal states in normal modes. This is widely used in continuous-variable quantum information [14].

2.3 Positive Maps, Complete Positivity and Kraus Representation

When describing open-system dynamics, we view the evolution as a linear map Λ on density matrices. Not every linear map is physically allowed.

2.3.1 Positive and Completely Positive Maps

Definition 2.3.1 — Positive Map. A linear map $\Lambda : \mathscr{B}(\mathscr{H}) \to \mathscr{B}(\mathscr{H})$ is *positive* if for all $\rho \geq 0$,

$$\rho \geq 0 \;\Rightarrow\; \Lambda(\rho) \geq 0.$$

Definition 2.3.2 — Completely Positive (CP) Map. A map Λ is completely positive if

$$\Lambda \otimes I_n$$

is positive for all $n \in \mathbb{N}$, where I_n is the identity map on $\mathscr{B}(\mathbb{C}^n)$.

R Complete positivity is stronger than positivity and is required because the system may be entangled with an ancilla. The extended map must not send a joint density matrix to a non-positive operator.

■ **Example 2.2 — Transpose Map.** The transpose map $T(\rho) = \rho^T$ is positive but not CP. If applied locally to one subsystem of an entangled state, it can produce a non-positive operator. This is used in Peres PPT entanglement criterion [10]. ■

2.3.2 Kraus Representation and CPTP Maps

Definition 2.3.3 — CPTP Map. A map Λ is a physically allowed quantum channel if it is completely positive and trace-preserving (CPTP):

$$\mathrm{Tr}(\Lambda(\rho)) = \mathrm{Tr}(\rho), \quad \forall \rho.$$

Theorem 2.3.1 — Kraus Representation. A map Λ is CPTP iff there exists a collection of operators $\{K_k\}$ such that

$$\Lambda(\rho) = \sum_k K_k \rho K_k^\dagger,$$

and

$$\sum_k K_k^\dagger K_k = I.$$

Sketch. One direction: Given such a representation, Λ is manifestly CP, because $\Lambda \otimes I$ has a similar Kraus form. Trace preservation follows from

$$\mathrm{Tr}(\Lambda(\rho)) = \sum_k \mathrm{Tr}(K_k \rho K_k^\dagger) = \mathrm{Tr}\left(\rho \sum_k K_k^\dagger K_k\right) = \mathrm{Tr}(\rho).$$

The converse uses Stinespring dilation: any CP map is obtained by unitary evolution on a larger system plus environment, followed by a partial trace. Expanding in an orthonormal basis of the environment yields Kraus operators [9, 1]. ■

R Kraus operators are not unique. Different sets of $\{K_k\}$ can define the same map. Unitary mixing of Kraus operators leaves the channel invariant.

2.4 Schmidt Decomposition and Entanglement in Bipartite Systems

Entanglement is the key resource and also the key "victim" of decoherence. We start with pure bipartite states.

2.4.1 Schmidt Decomposition

Let $|\Psi\rangle$ be a pure state in $\mathscr{H}_A \otimes \mathscr{H}_B$.

Theorem 2.4.1 — Schmidt Decomposition. There exist orthonormal bases $\{|e_k\rangle_A\}$ of $\mathscr{H}_A$ and $\{|f_k\rangle_B\}$ of $\mathscr{H}_B$ such that

$$|\Psi\rangle = \sum_{k=1}^{r} \sqrt{\lambda_k}\, |e_k\rangle_A \otimes |f_k\rangle_B,$$

with $\lambda_k \geq 0$, $\sum_k \lambda_k = 1$ and $r \leq \min(\dim \mathscr{H}_A, \dim \mathscr{H}_B)$.

Sketch. Write $|\Psi\rangle = \sum_{ij} C_{ij} |i\rangle_A \otimes |j\rangle_B$. Treat C as a matrix and perform its singular value decomposition $C = U\Sigma V^\dagger$, where Σ is diagonal. Define new bases $|e_k\rangle_A = \sum_i U_{ik}|i\rangle_A$, $|f_k\rangle_B = \sum_j V^*_{jk}|j\rangle_B$. The diagonal entries of Σ are $\sqrt{\lambda_k}$. ■

The λ_k are called *Schmidt coefficients*. The Schmidt rank r quantifies entanglement.

R If and only if $r = 1$, the state is separable (product state). Any $r > 1$ pure bipartite state is entangled.

2.4.2 Reduced Density Matrices and Entanglement Entropy

The reduced density matrix of subsystem A is

$$\rho_A = \mathrm{Tr}_B |\Psi\rangle\langle\Psi| = \sum_k \lambda_k |e_k\rangle\langle e_k|.$$

Definition 2.4.1 — Entanglement Entropy. The entanglement entropy of a pure bipartite state is the von Neumann entropy of either reduced density matrix:

$$S(\rho_A) = -\mathrm{Tr}(\rho_A \log \rho_A) = -\sum_k \lambda_k \log \lambda_k.$$

■ **Example 2.3 — Bell State.**

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big).$$

Schmidt coefficients are $\lambda_1 = \lambda_2 = 1/2$, so $S(\rho_A) = \log 2$, which is maximal for a $2 \otimes 2$ system. ■

2.4.3 Mixed States and PPT Criterion (Brief)

For mixed bipartite states ρ_{AB}, separability is defined as

$$\rho_{AB} \text{ separable} \iff \rho_{AB} = \sum_k p_k \rho_A^{(k)} \otimes \rho_B^{(k)}, \quad p_k \geq 0, \sum_k p_k = 1.$$

A necessary condition for separability is positivity under partial transpose (PPT) [10].

Definition 2.4.2 — Partial Transpose. In a product basis $|i\rangle_A \otimes |j\rangle_B$, define

$$\left(\rho^{T_B}\right)_{ij,kl} = \rho_{il,kj}.$$

Theorem 2.4.2 — PeresHorodecki Criterion. If ρ_{AB} is separable then $\rho_{AB}^{T_B} \geq 0$. For $2 \otimes 2$ and $2 \otimes 3$ systems the converse also holds.

R PPT is easy to test numerically and provides a powerful entanglement witness, especially in low dimensions.

2.5 Two-Level Systems and Bloch Sphere Representation

We now specialize to the simplest nontrivial quantum system: a qubit. Two-level systems are ubiquitous (spin-1/2, two-level atoms, superconducting qubits) and form the basic building blocks of quantum information and decoherence models.

2.5.1 Pauli Matrices and Operator Basis

The Pauli matrices are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Along with the identity I, they form a basis for 2×2 matrices.

Proposition 2.5.1 Any qubit density matrix ρ can be written as

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma}),$$

for some real vector $\vec{r} = (r_x, r_y, r_z)$ with $|\vec{r}| \leq 1$.

Sketch. Express ρ as a Hermitian matrix with unit trace,

$$\rho = \begin{pmatrix} a & c \\ c^* & 1-a \end{pmatrix}, \quad 0 \leq a \leq 1.$$

Then set $r_z = 2a - 1$, $r_x = 2\mathrm{Re}(c)$, $r_y = -2\mathrm{Im}(c)$, and check that $\rho = \frac{1}{2}(I + r_x\sigma_x + r_y\sigma_y + r_z\sigma_z)$. Positivity implies $|\vec{r}| \leq 1$. ■

Definition 2.5.1 — Bloch Sphere. The unit sphere in $\mathbb{R}^3$ with coordinates $\vec{r}$ is called the Bloch sphere. Pure states correspond to $|\vec{r}| = 1$, mixed states lie in the interior.

R The Bloch sphere provides a complete geometric description of qubit states. Quantum channels map the Bloch ball to itself, often as affine transformations (shrinking, rotating, shifting).

2.5.2 Parametrization of Pure States

Every pure qubit state can be written as

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle,$$

with $0 \leq \theta \leq \pi$, $0 \leq \phi < 2\pi$.

Proposition 2.5.2 The corresponding Bloch vector is

$$r_x = \sin\theta\cos\phi, \quad r_y = \sin\theta\sin\phi, \quad r_z = \cos\theta.$$

Proof. Compute $\langle\sigma_i\rangle = \langle\psi|\sigma_i|\psi\rangle$ and match with $\vec{r}\cdot\vec{\sigma}$ representation. It is straightforward algebra. ■

Hence the Bloch sphere is simply the unit sphere in (r_x, r_y, r_z).

2.5.3 Unitary Dynamics as Rotations on the Bloch Sphere

A general qubit Hamiltonian can be written as

$$H = \frac{\hbar}{2}\vec{\omega}\cdot\vec{\sigma},$$

where $\vec{\omega}$ is a real vector. The unitary time evolution is

$$U(t) = e^{-iHt/\hbar} = e^{-i(\vec{\omega}\cdot\vec{\sigma})t/2}.$$

Theorem 2.5.3 Under $U(t)$, the Bloch vector $\vec{r}(0)$ evolves by a rotation:

$$\vec{r}(t) = R_{\vec{\omega}}(t)\vec{r}(0),$$

where $R_{\vec{\omega}}(t)$ is a rotation matrix in $\mathrm{SO}(3)$ about axis $\vec{\omega}$ by angle $|\vec{\omega}|t$.

R Thus unitary dynamics of a qubit is geometrically just rotation of the Bloch vector. Control pulses in quantum computing are nothing but designed rotations on the Bloch sphere.

2.5.4 Bloch Sphere Picture of Simple Noise Channels

Open system decoherence often shrinks or distorts the Bloch ball.

Phase damping channel (dephasing).
In the computational basis $\{|0\rangle, |1\rangle\}$, phase damping suppresses off-diagonal elements:

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \longrightarrow \rho' = \begin{pmatrix} \rho_{00} & e^{-\Gamma t}\rho_{01} \\ e^{-\Gamma t}\rho_{10} & \rho_{11} \end{pmatrix}.$$

On the Bloch sphere, this is

$$r_x(t) = e^{-\Gamma t}r_x(0), \quad r_y(t) = e^{-\Gamma t}r_y(0), \quad r_z(t) = r_z(0).$$

R Phase damping leaves populations (r_z) invariant but shrinks coherence (r_x, r_y), visualizing decoherence as a contraction towards the z-axis.

Amplitude damping channel.
Models energy relaxation from $|1\rangle$ to $|0\rangle$ with probability γ:

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}.$$

The mapping of the Bloch vector is affine:

$$r'_x = \sqrt{1-\gamma}\, r_x, \quad r'_y = \sqrt{1-\gamma}\, r_y, \quad r'_z = (1-\gamma) r_z - \gamma.$$

R The Bloch sphere is contracted towards the south pole ($|0\rangle$), representing spontaneous emission and thermalization to the ground state.

2.5.5 Simple Bloch Sphere Diagram

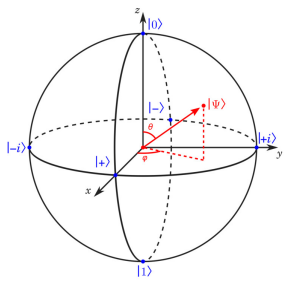


Figure 2.1: Geometric representation of a qubit state on the Bloch sphere. Pure states lie on the surface; mixed states lie inside the ball.

2.6 Summary

In this chapter we have developed a unified mathematical framework for describing quantum states in ways particularly suited to decoherence and open quantum systems:

- Phase-space formulations (Wigner, P, Q) provide quasi-probability views of quantum states and reveal classical limits and non-classical features such as Wigner negativity.
- Covariance matrix and Gaussian state formalism allow compact treatment of continuous-variable systems and harmonic baths, with Williamsons theorem providing normal forms.
- Positive and completely positive maps (CP, CPTP) describe physically allowed quantum operations and channels; Kraus representation gives us a powerful tool for analysis of open-system dynamics.
- Schmidt decomposition fully characterizes pure-state entanglement in bipartite systems, leading to entanglement entropy and PPT criteria for mixed states.
- Two-level systems (qubits) admit a complete geometric description on the Bloch sphere, where unitary dynamics are rotations and decoherence channels are affine contractions of the Bloch ball.

These tools will be used heavily in subsequent chapters to derive master equations, to quantify decoherence and dissipation, and to analyze how classical behavior emerges from underlying quantum dynamics.

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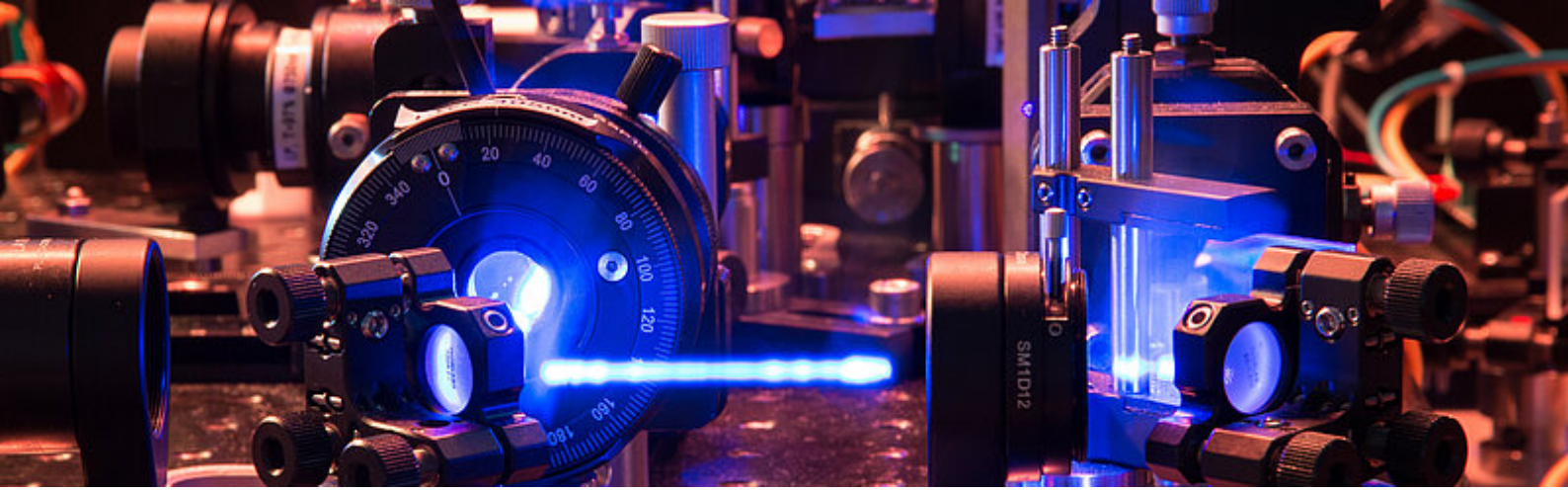


3. Introduction to Open Quantum Systems

Quantum systems interacting with large, uncontrolled environments exhibit irreversible behaviour such as relaxation, decoherence and dissipation. A complete description of such dynamics requires the framework of *open quantum systems* [1, 13]. This chapter develops Markovian master equations beginning with the classical Fokker–Planck picture, followed by their quantum generalisations, and then derives explicit reduced dynamics for several paradigmatic models. Rigorous mathematical statements are complemented by intuitive commentary, so that each derivation is physically motivated and mathematically transparent.

3.1 Classical Stochastic Evolution: From Liouville to Fokker–Planck

Definition 3.1.1 — Phase-space probability density. A classical system with phase-space coordinates $(x,p) \in \mathbb{R}^2$ is described statistically by a probability density $\rho(x,p,t)$ satisfying $\rho \geq 0$ and $\int dx\,dp\,\rho = 1$.

Theorem 3.1.1 — Liouville equation. For Hamiltonian dynamics $\dot{x} = \partial_p H$, $\dot{p} = -\partial_x H$, the probability density evolves as

$$\frac{\partial \rho}{\partial t} = -\{H,\rho\} := -\partial_x H\,\partial_p \rho + \partial_p H\,\partial_x \rho.$$

This equation is reversible, volume-preserving and deterministic. To model *noise* and *irreversibility*, one supplements Hamiltonian motion with diffusion and damping terms.

Definition 3.1.2 — Fokker–Planck equation. Let the dynamics include random forces. The probability density satisfies a second-order differential equation

$$\frac{\partial \rho}{\partial t} = -\sum_i \frac{\partial}{\partial x_i}\left[A_i(x)\rho\right] + \frac{1}{2}\sum_{ij} \frac{\partial^2}{\partial x_i \partial x_j}\left[D_{ij}(x)\,\rho\right]. \tag{3.1}$$

Here A_i is the drift vector and D_{ij} the diffusion matrix.

R The Fokker–Planck equation is the classical prototype of a Markovian master equation: first-order in time, linear, and preserving probabilities. Quantum master equations will be the noncommutative generalisation of this structure.

3.2 System–Bath Decomposition and Reduced Dynamics

In quantum mechanics, the state of the combined system $S+E$ ("environment" or "bath") is described by a density operator ρ_{SE} on $\mathscr{H}_S \otimes \mathscr{H}_E$. Only measurements on S are accessible, so we define:

Definition 3.2.1 — Reduced state. The state of the system alone is obtained via the partial trace:

$$\rho_S(t) = \mathrm{Tr}_E\, \rho_{SE}(t).$$

Time evolution of the total system is unitary:

$$\rho_{SE}(t) = U(t)\,\rho_{SE}(0)\,U^\dagger(t), \qquad U(t) = e^{-iHt/\hbar}.$$

The task of open-system theory is to obtain a closed equation of motion for $\rho_S(t)$.

3.3 Born, Markov and Coarse-Graining Approximations

Let $H = H_S + H_E + H_I$ with interaction

$$H_I = \sum_\alpha A_\alpha \otimes B_\alpha.$$

Definition 3.3.1 — Born approximation. Assume weak system–environment coupling so that correlations remain small:

$$\rho_{SE}(t) \approx \rho_S(t) \otimes \rho_E.$$

The environment is assumed stationary (e.g., thermal state).

Definition 3.3.2 — Markov approximation. Assume bath correlation functions decay on a timescale τ_B much shorter than the system evolution time τ_S:

$$\tau_B \ll \tau_S.$$

Thus the environment has no memory, enabling the replacement $\rho_S(t') \to \rho_S(t)$ under the integral kernel in the exact equation.

R Technically, the exact reduced dynamics is given by projection-operator formalisms: the Nakajima–Zwanzig equation (with memory kernel) and the time-convolutionless (TCL) equation (time-local). Born–Markov approximations eliminate the memory kernel, yielding a first-order differential equation.

Proposition 3.3.1 — Born–Markov master equation, interaction picture. Under these approximations,

$$\frac{d}{dt}\rho_S(t) = -\frac{1}{\hbar^2}\int_0^\infty d\tau\, \mathrm{Tr}_E\left[H_I(t), \left[H_I(t-\tau), \rho_S(t)\otimes\rho_E\right]\right].$$

The final step is the rotating-wave (secular) approximation, which discards fast-oscillating terms.

3.4 Gorini-Kossakowski-Sudarshan-Lindblad Equation

Theorem 3.4.1 — GKSL Master Equation. Any Markovian, completely positive and trace-preserving quantum dynamical semigroup $\{\Phi_t\}_{t\geq 0}$ admits a generator of the form

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H_{\text{eff}}, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right), \tag{3.2}$$

where H_{eff} is Hermitian and L_k are Lindblad (noise) operators.

R The term $L_k \rho L_k^\dagger$ describes "quantum jumps" while the anticommutator gives continuous dissipation. The semigroup property $\Phi_{t+s} = \Phi_t \circ \Phi_s$ is the quantum analogue of the classical Markov property.

3.5 Model I: Harmonic Oscillator Coupled to an Oscillator Bath

Let the system be a harmonic oscillator:

$$H_S = \hbar\omega a^\dagger a,$$

coupled linearly to a bath of oscillators:

$$H_I = (a + a^\dagger) \sum_k g_k (b_k + b_k^\dagger).$$

Definition 3.5.1 — Spectral density.

$$J(\omega) = \sum_k |g_k|^2 \, \delta(\omega - \omega_k)$$

characterises the bath.

Applying the Born–Markov + RWA approximations yields the standard quantum optical ME:

Theorem 3.5.1 — Master equation for quantum Brownian motion.

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[\omega a^\dagger a, \rho] + \gamma(\bar{n}+1)\left(a\rho a^\dagger - \tfrac{1}{2}\{a^\dagger a, \rho\}\right) + \gamma\bar{n}\left(a^\dagger \rho a - \tfrac{1}{2}\{aa^\dagger, \rho\}\right),$$

where γ is the damping rate and $\bar{n} = (e^{\hbar\omega/k_B T} - 1)^{-1}$.

R The first dissipation channel describes spontaneous and stimulated emission, the second describes absorption from the thermal bath. The stationary state is the thermal Gibbs state.

3.6 Model II: The Jaynes–Cummings Model (Closed + Dissipative)

Closed JCM

$$H_{\text{JCM}} = \hbar\omega\, a^\dagger a + \frac{\hbar\omega_0}{2}\sigma_z + \hbar g(\sigma_+ a + \sigma_- a^\dagger).$$

Theorem 3.6.1 — JCM eigenstates (dressed states). For each photon number n, the doublet subspace $\{|e,n\rangle, |g,n+1\rangle\}$ has eigenstates

$$|\pm,n\rangle = \cos\theta_n\,|e,n\rangle \pm \sin\theta_n\,|g,n+1\rangle,$$

with splitting $2\hbar\Omega_n$ where $\Omega_n = \sqrt{g^2(n+1)+(\Delta/2)^2}$.

R JCM predicts Rabi oscillations, collapses and revivals when the field has Poisson photon-number fluctuations.

Dissipative JCM (cavity loss)

Let κ be the cavity decay rate. The combined master equation is

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H_{\mathrm{JCM}},\rho] + \kappa\left(a\rho a^\dagger - \frac{1}{2}\{a^\dagger a,\rho\}\right). \tag{3.3}$$

R This is a minimal model combining unitary atomfield coupling with photon leakage, leading to damping of Rabi oscillations.

3.7 Model III: Two-Level System Coupled to a Spin Bath

Consider

$$H_I = \sigma_z \otimes \sum_k g_k\,\sigma_z^{(k)},$$

with bath spins initially in thermal equilibrium.

Theorem 3.7.1 — Pure dephasing ME from spin bath. Under Born–Markov approximations,

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H_S,\rho] - \Gamma_\varphi\left(\sigma_z\rho\sigma_z - \rho\right),$$

where the dephasing rate is $\Gamma_\varphi = \sum_k g_k^2 S_{zz}^{(k)}(0)$.

R There is *no* term that changes populations: the model conserves σ_z. Only the off-diagonal coherence decays.

3.8 Model IV: Phase-Damping Channel

Definition 3.8.1 — Dephasing channel.

$$\mathscr{E}(\rho) = (1-p)\rho + p\,\sigma_z\rho\sigma_z.$$

Proposition 3.8.1 — Lindblad form. The generator is

$$\frac{d\rho}{dt} = \gamma\left(\sigma_z\rho\sigma_z - \rho\right), \qquad \gamma = p/t.$$

R In the Bloch representation $\rho = \frac{1}{2}(I + \vec{r}\cdot\vec{\sigma})$, the x and y components decay as $e^{-2\gamma t}$ while r_z is invariant.

3.9 Gaussian States, Covariance Matrices and Wigner Functions

Gaussian states are central in quantum optics and open-system theory [14].

Definition 3.9.1 — Canonical quadratures.

$$R = (x,p)^T, \qquad x = \frac{a + a^\dagger}{\sqrt{2}}, \quad p = \frac{a - a^\dagger}{i\sqrt{2}}.$$

Definition 3.9.2 — Covariance matrix.

$$V_{ij} = \frac{1}{2}\langle R_i R_j + R_j R_i\rangle - \langle R_i\rangle\langle R_j\rangle.$$

Theorem 3.9.1 — Robertson–Schrödinger uncertainty.

$$V + \frac{i}{2}\Omega \geq 0, \qquad \Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Definition 3.9.3 — Wigner function. For density operator ρ,

$$W(x,p) = \frac{1}{\pi}\int_{-\infty}^{\infty} dy\, e^{2ipy}\,\langle x - y|\rho|x + y\rangle.$$

Theorem 3.9.2 — Gaussian Wigner function. If ρ is Gaussian with mean $\bar{R}$ and covariance V, then

$$W(R) = \frac{1}{2\pi\sqrt{\det V}}\exp\left[-\frac{1}{2}(R - \bar{R})^T V^{-1}(R - \bar{R})\right].$$

Wigner function of a coherent state

Let $|\alpha\rangle$ be a coherent state with $x_0 = \sqrt{2}\,\mathrm{Re}(\alpha)$, $p_0 = \sqrt{2}\,\mathrm{Im}(\alpha)$.

Proposition 3.9.3

$$W_\alpha(x,p) = \frac{1}{\pi}\exp\left[-(x - x_0)^2 - (p - p_0)^2\right].$$

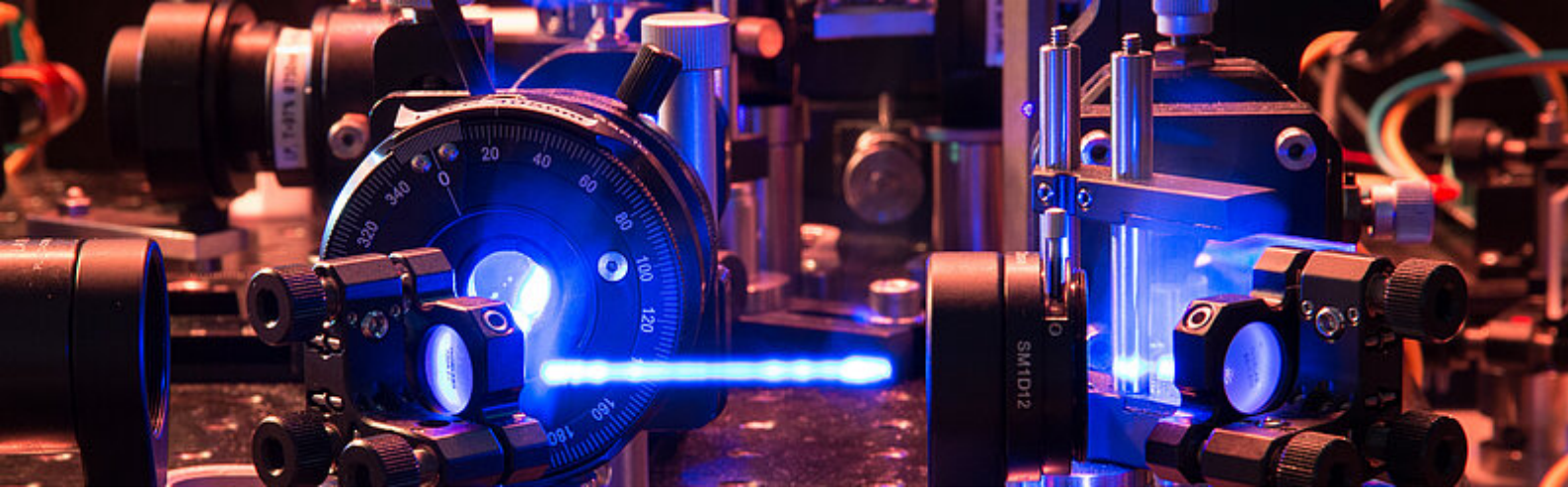
R The covariance matrix equals $V = \frac{1}{2}I$, the minimal-uncertainty case.

3.10 Summary

- Classical Liouville dynamics generalises to noisy evolution via the Fokker–Planck equation.
- Born and Markov approximations allow derivation of memoryless master equations.
- The GKSL theorem characterises all trace-preserving, completely positive Markovian evolutions.
- Canonical models (HO bath, JCM, spin bath, dephasing) illustrate dissipation and decoherence mechanisms.

- Gaussian states admit compact descriptions using moments and covariance matrices; Wigner functions describe their phase-space structure.

This framework forms the mathematical foundation for the study of decoherence, dissipation, quantum noise and irreversible dynamics in complex quantum systems.



4. Summary and Conceptual Overview of DOQS

Open quantum systems provide the mathematical and physical framework for describing realistic quantum dynamics in the presence of environmental noise, dissipation and decoherence. This summary page distills the central ideas, equations, approximations, and example models developed in the preceding chapters. It is intended both as a conceptual overview and as a quick reference.

4.1 The System–Environment Paradigm

Definition 4.1.1 — Open quantum system. A quantum system S interacting with an inaccessible environment E is described by the tensor-product Hilbert space $\mathscr{H}_S \otimes \mathscr{H}_E$ and the total Hamiltonian

$$H = H_S + H_E + H_I.$$

Only degrees of freedom of S are observable; thus its state is the reduced density operator $\rho_S(t) = \mathrm{Tr}_E\big(\rho_{SE}(t)\big)$.

R Because the environment is traced out, $\rho_S(t)$ typically becomes mixed, even when the total state was initially pure. This loss of purity reflects decoherence and entanglement with E.

4.2 Classical Analogy: From Liouville to Fokker–Planck

Classical stochastic processes motivate the structure of quantum master equations.

- The **Liouville equation** describes Hamiltonian evolution of a PDF in phase space.
- Adding drift and diffusion yields the **Fokker–Planck equation**, the classical prototype of a Markovian, first-order, linear time evolution.

R Quantum master equations reproduce these features in the operator setting: linearity, trace preservation, semigroup structure, and positivity.

4.3 Approximations Leading to Markovian Dynamics

Definition 4.3.1 — Born approximation. Assume weak coupling such that system–environment correlations remain small, allowing the factorised form $\rho_{SE}(t) \approx \rho_S(t) \otimes \rho_E$.

Definition 4.3.2 — Markov approximation. Assume short bath correlation time τ_B relative to system timescale τ_S. This removes the memory kernel and yields a time-local evolution equation.

R The rigorous projection-operator approaches (Nakajima–Zwanzig equation, TCL equation) demonstrate that these approximations suppress long-time environmental memory.

4.4 Lindblad Structure and Quantum Dynamical Semigroups

Theorem 4.4.1 — GKSL master equation. Any completely positive, trace-preserving (CPTP) Markovian semigroup $\{\Phi_t\}$ has generator

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H_{\text{eff}}, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \right).$$

- The Hamiltonian part describes coherent evolution (plus Lamb-shift correction).
- Lindblad operators L_k generate irreversible processes (decay, dephasing, diffusion).
- The equation guarantees complete positivity of the reduced dynamics.

R The dissipative term consists of two parts: a "jump" term $L\rho L^\dagger$ and a "drift" term $-\frac{1}{2}\{L^\dagger L, \rho\}$. Together they conserve trace and ensure monotonic decay of coherences.

4.5 Canonical Open-System Models

1. Harmonic oscillator + thermal oscillator bath

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[\omega a^\dagger a, \rho] + \gamma(\bar{n}+1)\mathscr{D}[a]\rho + \gamma\bar{n}\mathscr{D}[a^\dagger]\rho,$$

where $\bar{n}$ is the BoseEinstein occupation and $\mathscr{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$.

R This model produces damping of the oscillator amplitude and approach to a thermal state.

2. Jaynes–Cummings model (with cavity damping)

Closed JCM gives Rabi oscillations; adding cavity loss yields

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H_{\text{JCM}}, \rho] + \kappa\mathscr{D}[a]\rho,$$

leading to damping of collapses and revivals.

3. Two-level system + spin bath

Pure dephasing model:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H_S, \rho] - \Gamma_\varphi(\sigma_z\rho\sigma_z - \rho).$$

R Populations remain constant; coherences decay exponentially.

4. Phase-damping channel

$$\frac{d\rho}{dt} = \gamma(\sigma_z \rho \sigma_z - \rho).$$

4.6 Gaussian States and Wigner Functions

Gaussian states are characterised entirely by first and second moments.

Definition 4.6.1 — Covariance matrix. For quadratures $R = (x, p)^T$,

$$V_{ij} = \tfrac{1}{2}\langle R_i R_j + R_j R_i \rangle - \langle R_i \rangle \langle R_j \rangle.$$

Their Wigner function is

$$W(R) = \frac{1}{2\pi\sqrt{\det V}} \exp\left[-\tfrac{1}{2}(R-\bar{R})^T V^{-1} (R-\bar{R})\right].$$

Coherent state

$$W_\alpha(x,p) = \frac{1}{\pi} \exp\left[-(x-x_0)^2 - (p-p_0)^2\right].$$

R Coherent states have minimal uncertainty $V = \frac{1}{2}I$ and follow classical trajectories under harmonic evolution.

4.7 Physical Mechanisms of Decoherence

- **Dephasing:** loss of coherence without change in populations.
- **Relaxation (amplitude damping):** exchange of energy with a bath.
- **Thermalisation:** approach to Gibbs state via detailed balance.
- **Diffusion:** spreading of Wigner function due to noise.
- **Dissipation:** damping of expectation values (e.g., $\langle a \rangle$).

R Decoherence converts superpositions into statistical mixtures by suppressing off-diagonal elements while preserving diagonal ones. The rate and form depend on systembath coupling structure.

4.8 Hierarchy of Approaches to Open Quantum Systems

- **Microscopic Hamiltonian approach:** derive ME from $H_S + H_E + H_I$.
- **Projection operator techniques:** NZ and TCL formalisms.
- **Markovian semigroup theory:** GKSL structure.
- **Gaussian-state phase-space methods:** Wigner, covariance matrices.
- **Quantum optical models:** JCM, cavity damping, spin baths.

4.9 One-Page Master Formula Sheet

Unitary evolution: $\rho(t) = U(t)\rho(0)U^\dagger(t)$.

Born approximation: $\rho_{SE}(t) \approx \rho_S(t) \otimes \rho_E$.

Markov approximation: short bath correlation time.

Lindblad form: $\dot{\rho} = -\frac{i}{\hbar}[H,\rho] + \sum_k \mathscr{D}[L_k]\rho$.

Jump operator term: $\mathscr{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$.

HO damping: $\dot{\rho} = \gamma(\bar{n}+1)\mathscr{D}[a]\rho + \gamma\bar{n}\mathscr{D}[a^\dagger]\rho$.

Dephasing: $\dot{\rho} = \gamma(\sigma_z\rho\sigma_z - \rho)$.

Gaussian Wigner: $W(R) = \dfrac{1}{2\pi\sqrt{\det V}} e^{-\frac{1}{2}(R-\bar{R})^T V^{-1}(R-\bar{R})}$.

4.10 Final Perspective

Open-system theory unifies classical stochastic dynamics with quantum coherence and measurement theory. The master equations derived under BornMarkov assumptions capture a wide range of realistic physical processes while remaining mathematically elegant and analytically tractable. Gaussian methods, Wigner functions and concrete microscopic models form a powerful toolkit for understanding decoherence, dissipation and irreversible quantum dynamics in modern quantum science.

III

Appendix

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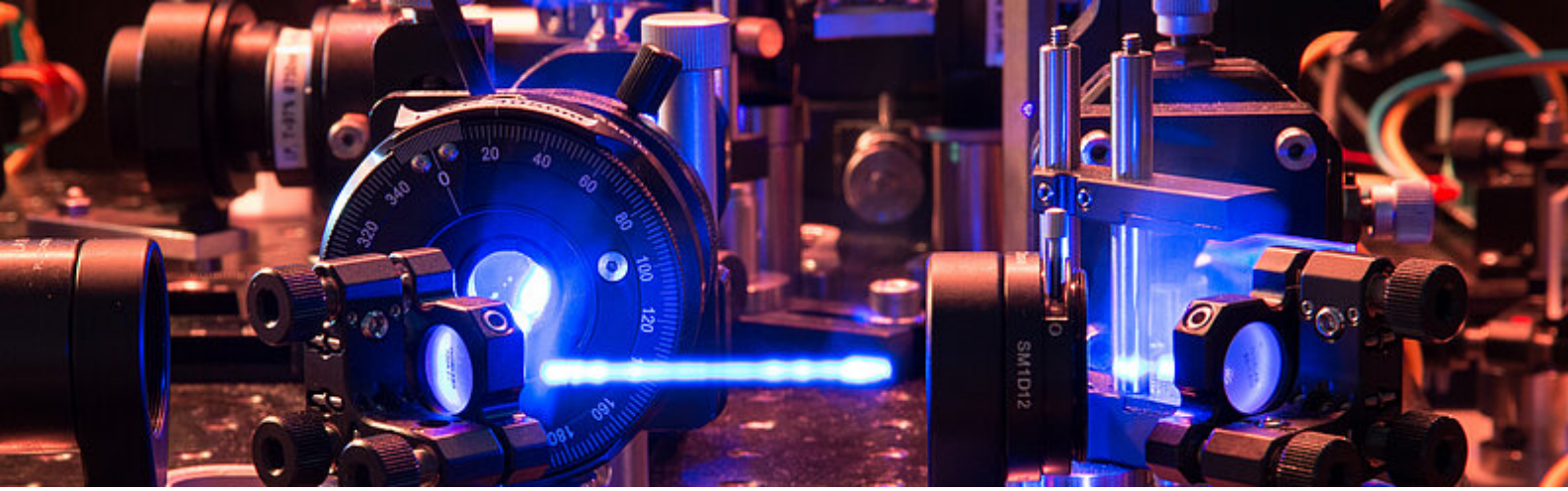
Books
Articles
Other Materials



Glossary of Terms

This glossary summarizes the major mathematical and physical terms used throughout these notes. Definitions are brief, cross-referenced, and grouped thematically.

A. Quantum Mechanics and Linear Algebra

Hilbert space $\mathscr{H}$: A complete inner-product space in which quantum states live.
Ket $|\psi\rangle$: A vector in Hilbert space representing a pure quantum state.
Density operator ρ: Positive semidefinite operator with unit trace encoding the statistical state (pure or mixed).
Trace Tr**:** Sum of diagonal matrix elements; used to compute expectation values.
Observable: Hermitian operator $A = A^\dagger$ whose eigenvalues are possible measurement outcomes.
Commutator [A,B :] Defined as $AB - BA$, measures incompatibility of observables.
Anti-commutator $\{A,B\}$: Defined as $AB + BA$, appears in Lindbladian generators.
Spectrum: Set of eigenvalues of an operator.
Rank: Number of nonzero eigenvalues of an operator.
Projector P: Operator satisfying $P^2 = P$; projects onto a subspace.
Unitary operator U: Operator preserving inner products: $U^\dagger U = I$.
Spectral decomposition: Representation of an operator as a weighted sum of projectors.

B. Density Matrices, Channels and Open Systems

Reduced state ρ_S: Partial trace over environment: $\rho_S = \mathrm{Tr}_E(\rho_{SE})$.
CPTP map: Completely positive, trace-preserving map describing physical evolution.
Kraus operators $\{E_k\}$: Operators such that $\mathscr{E}(\rho) = \sum_k E_k \rho E_k^\dagger$.
Lindblad operator L_k: Jump operator generating dissipative evolution.
Lindblad master equation:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H,\rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right).$$

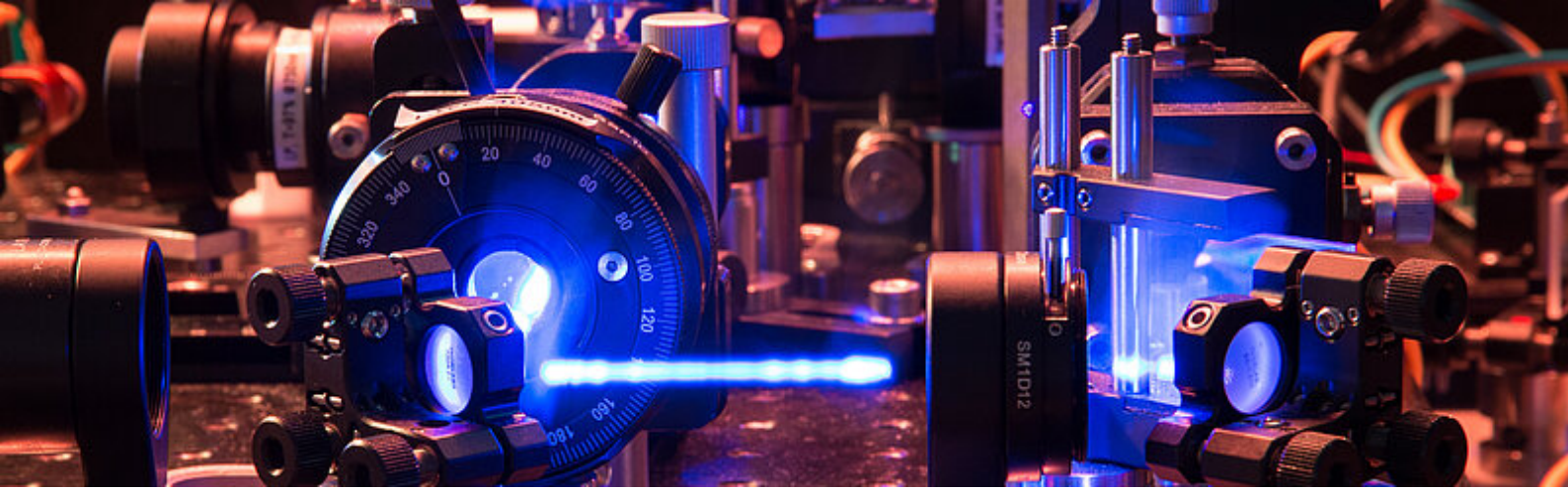
Dephasing: Loss of coherence without energy exchange.
Dissipation: Relaxation involving energy exchange with the environment.
Born approximation: Assumption of weak coupling and negligible system–environment correlations.
Markov approximation: Assumption of short environment memory time.
Non-Markovianity: Memory effects causing backflow of information.

C. Gaussian States and Phase Space

Quadratures (x, p)**:** Canonical observables of a mode, $[x, p] = i$.
Covariance matrix V**:** Matrix of second moments of quadratures.
Wigner function $W(x, p)$**:** Real quasi-probability distribution representing the state.
Gaussian state: State fully characterized by mean vector and covariance matrix.
Coherent state $|\alpha\rangle$**:** Minimum-uncertainty displaced vacuum state.
Squeezed state: State with reduced variance in one quadrature.

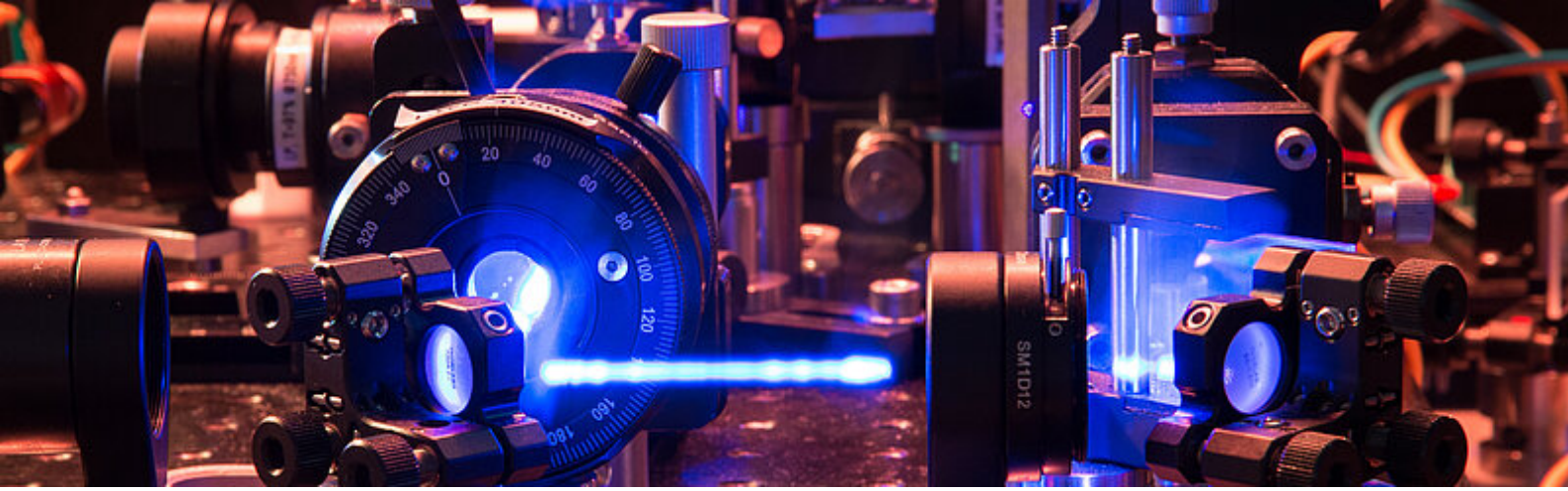
D. Common Models in Open Quantum Systems

Jaynes–Cummings model: A two-level atom interacting with a single cavity mode.
Spin-boson model: Two-level system coupled to many harmonic oscillators.
Quantum Brownian motion: Harmonic oscillator interacting with continuous bath of oscillators.
Phase-damping channel: A pure-dephasing completely positive map.
Amplitude-damping channel: Processes where excitations leak to the environment.



Notation Table

Symbol	Meaning
$\mathscr{H}$	Hilbert space of the system
$\lvert\psi\rangle$	Pure state vector
ρ	Density operator of a quantum state
Tr	Trace over a Hilbert space
Tr_E	Partial trace over environment
$\langle A\rangle = \mathrm{Tr}(\rho A)$	Expectation value of observable A
$\mathscr{E}$	Quantum channel (CPTP map)
$\{E_k\}$	Kraus operators of $\mathscr{E}$
L_k	Lindblad jump operators
$\mathscr{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$	Dissipator superoperator
$a, a^\dagger$	Annihilation and creation operators
$\sigma_x, \sigma_y, \sigma_z$	Pauli matrices
ω	Oscillator or transition frequency
γ, κ	Damping or decay rates
$R = (x, p)^T$	Quadrature vector
V	Covariance matrix of quadratures
$W(x, p)$	Wigner function
Ω	Symplectic form
S	Symplectic matrix implementing phase-space transformation
$\bar{n}$	Mean thermal occupation of a bosonic bath



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